



Jiamin Pan, Ph.D. Candidate

📍 Department of Mathematics & Statistics, Georgia State University

☎ +1 (404) 860-0680

✉ jpan4@gsu.edu

🆎 <https://orcid.org/0009-0009-3333-158X>

🌐 <https://www.linkedin.com/in/jiamin-pan-575083300/>

Education

01/2022 – 12/2026 (est.)

Ph.D. in Mathematics

Advisors: Dr. Florian Enescu

From Georgia State University, Atlanta, GA; GPA: 4.22/4.30 (Current)

07/06/2026 – 07/18/2026

SLMath Graduate Summer School Singularities in Commutative Algebra Through Cohomological Methods

Homepage: <https://www.slmath.org/summer-schools/1150>

Focuses on Lie algebra methods from homotopy theory, and support theoretic methods from the representation theory of finite groups.

08/2019 – 12/2021

M.S. in Science

Major: Mathematics; Concentration: Statistics;

From Georgia State University, Atlanta, GA; GPA: 4.18/4.30

08/2018 – 06/2019

Graduate Preparation Program in Engineering (GPP-E)

Major: Computer Science;

From University of California, Riverside, CA; GPA: 3.79/4.00

09/2015 – 06/2018

Bachelor of Engineering

Major: Software Engineering;

From Huazhong University of Science & Technology, Wuhan, Hubei, China;

GPA: 83.0/100 (3.59/4.00)

Research Interests

Commutative Algebra, Combinatorial Matrix Theory.

Research Publications and Preprints

- 1 M. Arav, F. J. Hall, H. van der Holst, Z. Li*, A. Mathivanan, J. Pan, H. Xu, and Z. Yang, Advances on similarity via transversal intersection of manifolds, *Linear Algebra and its Applications*, Volume 721 (2025), 102–121. doi: 10.1016/j.laa.2024.05.013
- 2 M. Arav, F. Hall, H. van der Holst, Z. Li*, Z.X. Li, J. Pan, H. Xu, Y. Xu, and Z. Yang, Orthogonal similarity and transversal intersection of manifolds, *Communications in Optimization Theory*, Vol. 2025 (2025), Article ID 27, pp. 1–21. doi: 10.23952/cot.2025.27
- 3 F. Enescu*, P. Kalla, J. Pan, Computing Boolean Functions from Polynomial Ideals with Prescribed Zeros, *Journal of Algebra and Its Applications* (2025). doi: 10.1142/S0219498826502762
- 4 M. Arav, F. Hall, Z. Li, H. van der Holst, C. Hart, Z.X. Li, J. Pan, J. Seo, L. Wang, H. Xu, Y. Xu, Z. Yang, and Y. Zhao, Matrices with the Rank-Preserving Transversality Property, Submitted to *Linear and Multilinear Algebra*.
- 5 F. Enescu, J. Pan, On the Finiteness of the Twisted Complexity for a Graded Ring in Prime Characteristic, *Bulletin Mathématique*, accepted for publication.

6 J. Pan, A Comprehensive Introduction to Algebra (Book), in preparation. Preview Link: <https://github.com/SolitaryHermit/alnotes>

Teaching Experiences

01/2021 – 12/2022
05/2023 – present

Graduate Teaching Assistant

Georgia State University. Courses Taught in Previous Semesters:

- MATH-0999: Support for College Algebra (Spring 2022);
- MATH-1001: Quantitative Reasoning (Spring 2021);
- MATH-1111: College Algebra (Fall 2021, Spring 2022, Fall 2022, Summer 2023);
- MATH-2211: Calculus of One Variable I (Summer 2024 - Lecture, Fall 2024 - Recitation);
- MATH-2212: Calculus of One Variable II - Recitations (Fall 2023, Spring 2024, Spring 2025, Fall 2025, Spring 2026, Fall 2026).

08/2019 – 12/2020
06/2026 – 07/2026

Graduate Laboratory Assistant

Georgia State University, University Commons MILE

- Tutoring MATH-1401: Elementary Statistics. (2019–2020)
- Tutoring MATH-1113: Precalculus. (2026)

Professional Experiences

06/2025 – 08/2025

AI Trainer – Mathematics Expert

The Handshake Model Validation Expert (MOVE) Fellowship.

- Developed and evaluated domain-specific prompts to assess the performance of large language models (LLMs) in mathematics.
- Analyzed LLM outputs for scientific accuracy, clarity, and depth in specialized subfields.
- Contributed to improving AI understanding of complex mathematics through expert review and feedback.
- Conducted independent research to support prompt development and evaluation tasks.

Research Program Application Reviewer

- Math Path Summer Program (05/2022);
- D-Map Summer Program (02/2025);
- Math Path Fall Assistantship (09/2025).

Talks and Presentations

- 1 Matrices with the Rank-Preserving Transversality Property, Departmental End-of-the-Year Celebration, Atlanta, GA, USA, 2025
- 2 Computing Boolean Functions from Polynomial Ideals with Prescribed Zeros, Commutative Algebra in The South, Atlanta, GA, USA, 2025
- 3 Properties of Uniformly F -Compatible Ideals, 105th Annual MAA-SE Section Meeting, Florence, AL, USA, 2026

Awards

- 1 V.V. Lavroff Graduate Award (\$700), 2025.
- 2 Wiley M. Suttles Math Award (\$520), 2024.
- 3 Wiley M. Suttles Math Award (\$500), 2023.
- 4 Best Graduate Teaching Assistant Award, 2023.

Technical skills

Programming Languages Python, R, Java, C, C++, Matlab, Singular, Macaulay2

Document Design $\LaTeX$, Basics of Markdown